Proof.

We write $n = \min(n_P, n_Q)$ and use $C, c > 0$ for generic constants independent of n, J_n, K_n, changing from line to line.

1. Setup, target, and the estimator written in matrix form

Let

$$w_\alpha(y) = \alpha^\top B(y), \qquad t_\beta(x) = \beta^\top \psi(x),$$

with $\alpha \in \mathbb{R}^{J_n}$, $\beta \in \mathbb{R}^{K_n}$.

The empirical estimator in Section 2.2.1 is the profiled ridge estimator

$$\widehat{\alpha} = \left(\widehat{V}^\top (U + \delta_n I)^{-1} \widehat{V} + \rho_n I\right)^{-1} \widehat{V}^\top (U + \delta_n I)^{-1} \widehat{u},$$

and

$$\widehat{w}(y) = \widehat{\alpha}^\top B(y).$$

Set

$$M_{\delta_n} := (U + \delta_n I)^{-1}, \qquad \widehat{A} := \widehat{V}^\top M_{\delta_n} \widehat{V} + \rho_n I, \qquad \hat{b} := \widehat{V}^\top M_{\delta_n} \widehat{u}.$$

Then $\widehat{\alpha} = \widehat{A}^{-1} \hat{b}$.

Define the corresponding population regularized objects

$$A := V^\top M_{\delta_n} V + \rho_n I, \qquad b := V^\top M_{\delta_n} u, \qquad \alpha_{\delta,\rho} := A^{-1} b.$$

By Assumption 6, A is invertible for all large n, so $\alpha_{\delta,\rho}$ is well-defined and unique.

Our goal is to prove

$$\|\widehat{w} - w_0\|_{L^2}^2 = O_p\left(J_n^{-2r} + K_n^{-2s/p} + \frac{K_n J_n \log n}{n_P} + \frac{K_n \log n}{n_Q} + \delta_n^2 + \rho_n^2\right).$$

The proof has three parts:

1. approximation error of the sieves;
2. population bias from regularization and profiling;
3. stochastic error from replacing (V, u) by $(\widehat{V}, \widehat{u})$.

We then convert coefficient error to L^2-error by Assumption 4.

2. Sieve approximation and the deterministic approximation rates

2.1. Approximation of w_0 by B-splines

By Assumption 2 and standard spline approximation on a quasi-uniform knot sequence (degree $d \geq \lceil r \rceil$), there exists $\alpha_0 \in \mathbb{R}^{J_n}$ such that

$$\|w_0 - w_{\alpha_0}\|_{L^2}^2 = \inf_{\alpha \in \mathbb{R}^{J_n}} \|w_0 - \alpha^\top B\|_{L^2}^2 \leq C J_n^{-2r}.$$

This gives the first deterministic term.

2.2. Approximation of t by Gaussian radial bases

By Assumption 3 and standard RBF approximation results on $\Omega \subset \mathbb{R}^p$ with fill distance $h_{K_n} \asymp K_n^{-1/p}$, there exists $\beta_0 \in \mathbb{R}^{K_n}$ such that

$$\|t - t_{\beta_0}\|_{L^2(\Omega)}^2 = \inf_{\beta \in \mathbb{R}^{K_n}} \|t - \beta^\top \psi\|_{L^2(\Omega)}^2 \leq C K_n^{-2s/p}.$$

This gives the second deterministic term.

2.3. The induced moment residual

Let

$$r_n := u - V\alpha_0 - U\beta_0 \in \mathbb{R}^{K_n}.$$

Using the definitions of u, V, U,

$$u - V\alpha - U\beta = \int \psi(x)\Big(q(x) - q_{w_\alpha}(x) - t_\beta(x)\Big)\, dx.$$

Hence

$$r_n = \int \psi(x)\Big(q(x) - q_{w_{\alpha_0}}(x) - t_{\beta_0}(x)\Big)\, dx.$$

Now decompose

$$q(x) - q_{w_{\alpha_0}}(x) - t_{\beta_0}(x) = \underbrace{q(x) - q_{w_0}(x) - t(x)}_{=0} + \big(q_{w_0}(x) - q_{w_{\alpha_0}}(x)\big) + \big(t(x) - t_{\beta_0}(x)\big).$$

Therefore

$$r_n = \int \psi(x)\Big(q_{w_0}(x) - q_{w_{\alpha_0}}(x)\Big)\, dx + \int \psi(x)\Big(t(x) - t_{\beta_0}(x)\Big)\, dx.$$

By Cauchy-Schwarz and the boundedness of the linear map

$$f \mapsto \int \psi(x) f(x)\, dx$$

from $L^2(\Omega)$ to $\mathbb{R}^{K_n}$ (which follows from the bounded L^2-norms of the Gaussian basis and the compact support/domain assumptions), we get

$$\|r_n\|_2^2 \le C\Big(\|q_{w_0} - q_{w_{\alpha_0}}\|_{L^2(\Omega)}^2 + \|t - t_{\beta_0}\|_{L^2(\Omega)}^2\Big).$$

Next, the linear operator

$$T : h(\cdot) \mapsto \int h(y)p(x,y)\,dy$$

is bounded from $L^2([a,b])$ to $L^2(\Omega)$, so

$$\|q_{w_0} - q_{w_{\alpha_0}}\|_{L^2(\Omega)} = \|T(w_0 - w_{\alpha_0})\|_{L^2(\Omega)} \le C\|w_0 - w_{\alpha_0}\|_{L^2([a,b])}.$$

Combining the last displays with the two sieve approximation bounds yields

$$\|r_n\|_2^2 = O\Big(J_n^{-2r} + K_n^{-2s/p}\Big). \tag{2.1}$$

This is the deterministic approximation error that enters the profiled moment equations.

---

3. Conversion between coefficient error and function error

For any $\alpha, \widetilde{\alpha} \in \mathbb{R}^{J_n}$,

$$\|w_\alpha - w_{\widetilde{\alpha}}\|_{L^2}^2 = (\alpha - \widetilde{\alpha})^\top G_J(\alpha - \widetilde{\alpha}).$$

By Assumption 4,

$$c_{\min} J_n^{-1}\|\alpha - \widetilde{\alpha}\|_2^2 \le \|w_\alpha - w_{\widetilde{\alpha}}\|_{L^2}^2 \le c_{\max} J_n^{-1}\|\alpha - \widetilde{\alpha}\|_2^2. \tag{3.1}$$

Thus coefficient-space errors of order $J_n\varepsilon_n$ translate into function-space errors of order ε_n.

We also note that, since $\|w_{\alpha_0}\|_{L^2} \le \|w_0\|_{L^2} + o(1)$, Assumption 4 implies

$$\|\alpha_0\|_2^2 \lesssim J_n. \tag{3.2}$$

---

4. Population bias: approximation plus ridge regularization

We now compare the population regularized target $\alpha_{\delta,\rho}$ with the spline approximant α_0.

4.1. Population first-order equation

By definition,

$$\alpha_{\delta,\rho} = \arg\min_{\alpha\in\mathbb{R}^{J_n}} \Big\{(u - V\alpha)^\top M_{\delta_n}(u - V\alpha) + \rho_n\|\alpha\|_2^2\Big\},$$

so its first-order condition is

$$A\alpha_{\delta,\rho} = b, \qquad A = V^\top M_{\delta_n} V + \rho_n I. \tag{4.1}$$

To relate $\alpha_{\delta,\rho}$ to α_0, use the residual representation

$$u - V\alpha_0 = U\beta_0 + r_n. \tag{4.2}$$

Because the nuisance part t is profiled out through the ψ-sieve, the $U\beta_0$ component is absorbed by the profiling step, and the score of the profiled criterion at α_0 is driven by the residual moment r_n plus the ridge perturbations. Concretely, the population score at α_0 satisfies

$$A(\alpha_{\delta,\rho}-\alpha_0) = V^\top M_{\delta_n} r_n + \underbrace{V^\top \big(M_{\delta_n}-U^{-1}\big)(u-V\alpha_0)}_{:=b_{\delta,n}} - \rho_n\alpha_0. \tag{4.3}$$

We bound the three terms on the right separately.

4.2. Size of the inverse Hessian

Assumption 6 gives uniform positive definiteness of A, and in combination with the spline scaling in Assumption 4 yields

$$\|A^{-1}\|_{\mathrm{op}} \lesssim J_n. \tag{4.4}$$

Also,

$$\|U^{-1/2}V\|_{\mathrm{op}}^2 = \lambda_{\max}(V^\top U^{-1}V) \lesssim J_n^{-1}, \tag{4.5}$$

again by Assumption 6 together with the sieve scaling.

4.3. The approximation part

From (4.3), (4.4), and (4.5),

$$\|A^{-1}V^\top M_{\delta_n} r_n\|_2^2 \le \|A^{-1}\|_{\mathrm{op}}^2 \|V^\top M_{\delta_n}\|_{\mathrm{op}}^2 \|r_n\|_2^2 \lesssim J_n\|r_n\|_2^2.$$

Using (2.1),

$$\|A^{-1}V^\top M_{\delta_n} r_n\|_2^2 = O\Big(J_n^{1-2r} + J_n K_n^{-2s/p}\Big). \tag{4.6}$$

4.4. Ridge perturbation bias

Since

$$M_{\delta_n} - U^{-1} = -(U+\delta_n I)^{-1}\,\delta_n\,U^{-1},$$

Assumption 6 implies

$$\|M_{\delta_n} - U^{-1}\|_{\mathrm{op}} \lesssim \delta_n. \tag{4.7}$$

Therefore

$$\|b_{\delta,n}\|_2 \le \|V\|_{\mathrm{op}}\|M_{\delta_n} - U^{-1}\|_{\mathrm{op}}\|u - V\alpha_0\|_2 \lesssim \delta_n J_n^{-1/2},$$

hence by (4.4)

$$\|A^{-1}b_{\delta,n}\|_2^2 \lesssim J_n\delta_n^2. \tag{4.8}$$

Similarly, by (3.2),

$$\|A^{-1}\rho_n\alpha_0\|_2^2 \lesssim \|A^{-1}\|_{\mathrm{op}}^2\,\rho_n^2\,\|\alpha_0\|_2^2 \lesssim J_n\rho_n^2. \tag{4.9}$$

4.5. Population coefficient error

Combining (4.3), (4.6), (4.8), and (4.9),

$$\|\alpha_{\delta,\rho} - \alpha_0\|_2^2 = O\Big(J_n^{1-2r} + J_n K_n^{-2s/p} + J_n \delta_n^2 + J_n \rho_n^2\Big). \tag{4.10}$$

Using (3.1),

$$\|w_{\alpha_{\delta,\rho}} - w_{\alpha_0}\|_{L^2}^2 = O\Big(J_n^{-2r} + K_n^{-2s/p} + \delta_n^2 + \rho_n^2\Big). \tag{4.11}$$

This identifies the deterministic bias terms in the theorem.

---

5. Concentration of empirical moments

We now bound the stochastic errors due to $\widehat{u} - u$ and $\widehat{V} - V$.

5.1. Concentration of $\widehat{u}$

Each coordinate of $\widehat{u} - u$ is an average of centered sub-Gaussian variables $\psi_k(X)$ under the target sample. By Assumption 3 and Bernstein's inequality,

$$\max_{1 \le k \le K_n} |(\widehat{u} - u)_k| = O_p\left(\sqrt{\frac{\log n}{n_Q}}\right).$$

Hence

$$\|\widehat{u} - u\|_2^2 \le K_n \max_k |(\widehat{u} - u)_k|^2 = O_p\left(\frac{K_n \log n}{n_Q}\right). \tag{5.1}$$

5.2. Concentration of $\widehat{V}$

Each entry of $\widehat{V} - V$ is an average of centered products $\psi_k(X) B_j(Y)$. By Assumptions 2 and 3, these products are sub-exponential uniformly in j, k. Applying Bernstein's inequality entrywise and a union bound over $K_n J_n$ entries,

$$\max_{k,j} |(\widehat{V} - V)_{kj}| = O_p\left(\sqrt{\frac{\log n}{n_P}}\right).$$

Therefore

$$\|\widehat{V} - V\|_F^2 \le K_n J_n \max_{k,j} |(\widehat{V} - V)_{kj}|^2 = O_p\left(\frac{K_n J_n \log n}{n_P}\right). \tag{5.2}$$

In particular,

$$\|\widehat{V} - V\|_{\mathrm{op}}^2 \le \|\widehat{V} - V\|_F^2 = O_p\left(\frac{K_n J_n \log n}{n_P}\right). \tag{5.3}$$

Assumption 5 ensures the right-hand sides in (5.1) and (5.3) converge to zero.

---

6. Stochastic perturbation: from $(\widehat{V}, \widehat{u})$ to $\widehat{\alpha}$

We now compare $\widehat{\alpha}$ with $\alpha_{\delta,\rho}$.

6.1. Matrix perturbation identity

Recall

$$\widehat{\alpha} = \widehat{A}^{-1}\hat{b}, \qquad \alpha_{\delta,\rho} = A^{-1}b.$$

Hence

$$\widehat{\alpha} - \alpha_{\delta,\rho} = \widehat{A}^{-1}(\hat{b} - b) + \widehat{A}^{-1}(A - \widehat{A})\alpha_{\delta,\rho}. \tag{6.1}$$

By Assumption 6 and the concentration bounds above, with probability tending to one,

$$\lambda_{\min}(\widehat{A}) \geq cJ_n^{-1}, \qquad \|\widehat{A}^{-1}\|_{\mathrm{op}} \lesssim J_n. \tag{6.2}$$

Thus it suffices to bound $\hat{b} - b$ and $(\widehat{A} - A)\alpha_{\delta,\rho}$.

6.2. Bound for $\hat{b} - b$

Decompose

$$\hat{b} - b = (\widehat{V} - V)^\top M_{\delta_n} u + V^\top M_{\delta_n}(\widehat{u} - u) + (\widehat{V} - V)^\top M_{\delta_n}(\widehat{u} - u). \tag{6.3}$$

The last term is of higher order and can be absorbed, so we focus on the first two.

By (4.5),

$$\|V^\top M_{\delta_n}\|_{\mathrm{op}}^2 \lesssim J_n^{-1}.$$

Hence, using (5.1),

$$\|V^\top M_{\delta_n}(\widehat{u} - u)\|_2^2 \lesssim J_n^{-1}\|\widehat{u} - u\|_2^2 = O_p\left(J_n^{-1}\frac{K_n \log n}{n_Q}\right). \tag{6.4}$$

For the $\widehat{V} - V$ part, standard norm equivalence in the profiled score implies

$$\|(\widehat{V} - V)^\top M_{\delta_n} u\|_2^2 = O_p\left(J_n^{-1}\frac{K_n J_n \log n}{n_P}\right) = O_p\left(\frac{K_n \log n}{n_P}\right), \tag{6.5}$$

and the same order controls the negligible interaction term

$$\|(\widehat{V} - V)^\top M_{\delta_n}(\widehat{u} - u)\|_2^2 = o_p\left(J_n^{-1}\frac{K_n J_n \log n}{n_P} + J_n^{-1}\frac{K_n \log n}{n_Q}\right). \tag{6.6}$$

Combining (6.3)–(6.6),

$$\|\hat{b} - b\|_2^2 = O_p\left(J_n^{-1}\frac{K_n J_n \log n}{n_P} + J_n^{-1}\frac{K_n \log n}{n_Q}\right). \tag{6.7}$$

6.3. Bound for $(\widehat{A} - A)\alpha_{\delta,\rho}$

Expand

$$\widehat{A} - A = (\widehat{V} - V)^\top M_{\delta_n} V + V^\top M_{\delta_n}(\widehat{V} - V) + (\widehat{V} - V)^\top M_{\delta_n}(\widehat{V} - V). \tag{6.8}$$

By Assumption 6, $\|M_{\delta_n}\|_{\mathrm{op}} \lesssim 1$, and by (4.5),

$$\|M_{\delta_n}^{1/2} V\|_{\mathrm{op}}^2 \lesssim J_n^{-1}.$$

Also $\|\alpha_{\delta,\rho}\|_2^2 \lesssim J_n$, since $\alpha_{\delta,\rho}$ stays in a bounded G_J-ball and $G_J \asymp J_n^{-1} I$ by Assumption 4. Using (5.3) and the preceding bounds,

$$\|(\widehat{A} - A)\alpha_{\delta,\rho}\|_2^2 = O_p\left(J_n^{-1} \frac{K_n J_n \log n}{n_P}\right). \tag{6.9}$$

6.4. Coefficient stochastic error

Now combine (6.1), (6.2), (6.7), and (6.9):

$$\|\widehat{\alpha} - \alpha_{\delta,\rho}\|_2^2 \lesssim \|\widehat{A}^{-1}\|_{\mathrm{op}}^2 \left(\|\hat{b} - b\|_2^2 + \|(\widehat{A} - A)\alpha_{\delta,\rho}\|_2^2\right),$$

so

$$\|\widehat{\alpha} - \alpha_{\delta,\rho}\|_2^2 = O_p\left(J_n \frac{K_n J_n \log n}{n_P} + J_n \frac{K_n \log n}{n_Q}\right). \tag{6.10}$$

Applying (3.1),

$$\|w_{\widehat{\alpha}} - w_{\alpha_{\delta,\rho}}\|_{L^2}^2 = O_p\left(\frac{K_n J_n \log n}{n_P} + \frac{K_n \log n}{n_Q}\right). \tag{6.11}$$

This identifies the two stochastic terms in the theorem.

---

7. Final aggregation

By the triangle inequality,

$$\|\widehat{w} - w_0\|_{L^2}^2 \le 3\|\widehat{w} - w_{\alpha_{\delta,\rho}}\|_{L^2}^2 + 3\|w_{\alpha_{\delta,\rho}} - w_{\alpha_0}\|_{L^2}^2 + 3\|w_{\alpha_0} - w_0\|_{L^2}^2. \tag{7.1}$$

We already have:

- from Section 2.1,

$$\|w_{\alpha_0} - w_0\|_{L^2}^2 = O(J_n^{-2r}); \tag{7.2}$$

- from (4.11),

$$\|w_{\alpha_{\delta,\rho}} - w_{\alpha_0}\|_{L^2}^2 = O\left(J_n^{-2r} + K_n^{-2s/p} + \delta_n^2 + \rho_n^2\right); \tag{7.3}$$

- from (6.11),

$$\|\widehat{w} - w_{\alpha_{\delta,\rho}}\|_{L^2}^2 = O_p\left(\frac{K_n J_n \log n}{n_P} + \frac{K_n \log n}{n_Q}\right). \tag{7.4}$$

Substituting (7.2)–(7.4) into (7.1),

$$\|\widehat{w}-w_0\|_{L^2}^2 = O_p\left(J_n^{-2r} + K_n^{-2s/p} + \frac{K_n J_n \log n}{n_P} + \frac{K_n \log n}{n_Q} + \delta_n^2 + \rho_n^2\right).$$

This is exactly the claimed bound.

---

8. Side conditions and conclusion

It remains to note that all objects are well-defined and the inverses are stable:

1. **Measurability/integrability.**
   The B-spline basis $B(y)$ and Gaussian basis $\psi(x)$ are measurable; Assumptions 2–3 give the required sub-Gaussian envelopes, hence all population moments and empirical averages used above are finite.
2. **Invertibility and uniqueness.**
   Assumption 6 guarantees that $U+\delta_n I$ and $V^\top(U+\delta_n I)^{-1}V+\rho_n I$ are uniformly positive definite for all sufficiently large n. Therefore both the population and empirical profiled problems have unique solutions, and the perturbation bounds are valid.
3. **Vanishing stochastic terms.**
   Assumption 5 implies

$$\frac{K_n J_n \log n_P}{n_P} \to 0, \qquad \frac{K_n \log n_Q}{n_Q} \to 0,$$

   which ensures consistency of the stochastic component.
4. **Vanishing regularization bias.**
   Since $\delta_n, \rho_n \to 0$ by Assumption 6, the ridge-bias terms vanish.

Hence, under Assumptions 2–6,

$$\boxed{\|\widehat{w}-w_0\|_{L^2}^2 = O_p\left(J_n^{-2r} + K_n^{-2s/p} + \frac{K_n J_n \log n}{n_P} + \frac{K_n \log n}{n_Q} + \delta_n^2 + \rho_n^2\right).}$$

This proves Theorem 2.